🛒 Shopper Spectrum: Customer Segmentation and Product Recommendation System:

Project Overview

Shopper Spectrum is an E-Commerce Analytics project designed to analyze customer purchasing behavior and generate personalized product recommendations. The project leverages RFM (Recency, Frequency, Monetary) Analysis, K-Means Clustering, and Item-Based Collaborative Filtering to identify customer segments and recommend relevant products.

The primary objective is to help businesses improve customer retention, optimize marketing strategies, and enhance customer experience through data-driven insights.

Problem Statement

The e-commerce industry generates large volumes of transaction data daily. Understanding customer purchasing behavior is critical for improving customer engagement and increasing revenue.

This project aims to:

- Analyze online retail transaction data.
- Segment customers using RFM Analysis.
- Identify High-Value, Regular, Occasional, and At-Risk customers.
- Develop a product recommendation system using collaborative filtering techniques.
- Deploy an interactive Streamlit application for real-time predictions and recommendations.

Business Use Cases

- Customer Segmentation for Targeted Marketing Campaigns
- Personalized Product Recommendations
- Customer Retention and Loyalty Programs
- Inventory Planning and Demand Forecasting
- Data-Driven Decision Making

Dataset Information

Dataset

Online Retail Dataset

Dataset Features

Column	Description
InvoiceNo	Transaction Number
StockCode	Product Code
Description	Product Name
Quantity	Quantity Purchased
InvoiceDate	Date and Time of Purchase
UnitPrice	Product Price
CustomerID	Customer Identifier
Country	Customer Location

Data Preprocessing

The following preprocessing steps were performed:

- Removed records with missing CustomerID values.
- Removed cancelled invoices.
- Removed transactions with zero or negative quantities.
- Removed transactions with zero or negative prices.
- Converted InvoiceDate into datetime format.
- Created TotalAmount feature.

Dataset Summary

Metric	Count
Original Records	541,909
Cleaned Records	397,884
Removed Records	144,025
Unique Customers	4,338
Unique Products	3,877

Exploratory Data Analysis (EDA)

The following analyses were performed:

Country Analysis

Transaction volumes were analyzed across countries to identify major customer markets.

Top-Selling Products

Top-performing products were identified based on purchase quantities.

Purchase Trend Analysis

Monthly transaction trends were visualized to understand seasonal purchasing patterns.

Monetary Analysis

Customer spending behavior was analyzed using transaction-level and customer-level monetary distributions.

RFM Distribution Analysis

Recency, Frequency, and Monetary distributions were examined to understand customer purchasing behavior.

Customer Segmentation

RFM Feature Engineering

Three customer behavior metrics were calculated:

Recency

Number of days since the customer's most recent purchase.

Frequency

Number of transactions completed by a customer.

Monetary

Total amount spent by a customer.

Data Transformation

- Log Transformation applied to reduce skewness.
- StandardScaler applied for normalization.

Clustering Methodology

Algorithm Used

K-Means Clustering

Cluster Selection

The optimal number of clusters was determined using:

- Elbow Method
- Silhouette Score Analysis

Final Number of Clusters

4 Clusters

Customer Segment Labels

Segment	Description
High-Value	Frequent, recent, and high-spending customers
Regular	Consistent and steady purchasers
Occasional	Customers with lower purchasing frequency
At-Risk	Customers inactive for a long period

Segment Distribution

Segment	Customers
At-Risk	1,612
Regular	1,173
Occasional	837
High-Value	716

Product Recommendation System

Recommendation Approach

Item-Based Collaborative Filtering

Methodology

1. Created Customer-Product Purchase Matrix.
2. Calculated Product Similarity using Cosine Similarity.
3. Recommended Top 5 similar products for the selected product.

Recommendation Output

The system returns five highly similar products based on customer purchase behavior.

Model Evaluation

Clustering Evaluation

- Elbow Curve Analysis
- Silhouette Score Analysis

Recommendation Evaluation

- Cosine Similarity Matrix
- Product Similarity Heatmap

Streamlit Application

The project includes an interactive Streamlit web application with two modules.

Customer Segmentation Module

Inputs:

- Recency
- Frequency
- Monetary

Output:

- High-Value
- Regular
- Occasional
- At-Risk

Product Recommendation Module

Input:

- Product Name

Output:

- Top 5 Recommended Products

Project Structure

```
shopper-spectrum-ecommerce
│
├── app.py
```

```
├── README.md
├── requirements.txt
│
├── data
│   └── online_retail.csv
│
├── models
│   ├── feature_columns.pkl
│   ├── kmeans.pkl
│   ├── scaler.pkl
│   ├── segment_mapping.pkl
│   └── product_list.pkl
│
├── notebook
│   └── Shopper_Spectrum.ipynb
│
└── Screenshots
    ├── home_page1.png
    ├── home_page2.png
    ├── Customer_Segmentation.png
    ├── Customer_Segmentation_Graph.png
    ├── Product_Segmentation.png
    ├── Product_heatmap_similarity_matrix.png
    └── RFM_Distribution_Analysis.png
```

Note: product_similarity.pkl was excluded from GitHub because it exceeds GitHub's 100 MB file size limit. The recommendation model can be regenerated by running the notebook.

Project Screenshots

Home Page

- home_page1.png
- home_page2.png

Customer Segmentation

- Customer_Segmentation.png
- Customer_Segmentation_Graph.png

Product Recommendation

- Product_Segmentation.png

Similarity Matrix

- Product_heatmap_similarity_matrix.png

RFM Distribution Analysis

- RFM_Distribution_Analysis.png

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Streamlit
- Pickle

Future Improvements

- Deploy application on Streamlit Cloud
- Use Hybrid Recommendation Systems
- Add Real-Time Customer Analytics Dashboard

Conclusion

This project successfully implemented customer segmentation and product recommendation techniques using real-world e-commerce transaction data. RFM Analysis and K-Means Clustering enabled effective customer categorization, while Item-Based Collaborative Filtering generated meaningful product recommendations. The final Streamlit application provides an interactive interface for real-time customer segmentation and product recommendation, making it suitable for business intelligence and customer engagement use cases.